

TRAWSGOED, United Kingdom

WMO: 035030

Lat: 52.3442N

Lon: 3.9481W

Elev: 62

StdP: 100.58

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.7	(c) -3.1	(d) -6.0	(e) 2.3	(f) -1.5	(g) -4.6	(h) 2.6	(i) -0.4	(j) 11.4	(k) 5.0	(l) 10.4	(m) 6.3	(n) 0.5	(o) 130	(p) 0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	24.2	18.1	22.2	17.1	20.6	16.3	18.9	22.5	17.9	20.9	17.1	19.6	3.8	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	12.7	20.1	16.7	12.0	19.1	16.0	11.5	18.4	53.7	22.6	50.7	20.9	48.3	19.7	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.4	8.3	7.3	-7.1	28.1	2.1	2.0	-8.6	29.5	-9.9	30.7	-11.1	31.8	-12.6	33.3
			-6.8	21.4	1.9	1.9	-8.2	22.8	-9.3	24.0	-10.3	25.1	-11.7	26.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	5.3	5.4	6.4	8.3	11.0	13.7	15.4	15.1	13.4	11.0	7.8	5.8		
	DBStd	4.71	3.34	3.17	3.02	2.90	2.83	2.55	2.22	2.17	2.43	2.95	3.36	3.79		
	HDD10.0	718	146	131	115	66	20	2	0	0	2	23	79	134		
	HDD18.3	3078	403	362	369	301	226	140	97	103	148	226	315	388		
	CDD10.0	691	2	2	4	14	52	114	166	158	105	56	14	4		
	CDD18.3	11	0	0	0	0	1	2	4	2	1	0	0	0		
	CDH23.3	99	0	0	0	1	4	31	48	11	4	1	0	0		
	CDH26.7	13	0	0	0	0	0	4	7	1	0	0	0	0		
(14) Wind	WSAvg	2.9	3.5	3.5	3.3	3.1	3.0	2.6	2.5	2.4	2.4	2.9	2.9	3.3		
(15) Precipitation	PrecAvg	1174	119	84	91	73	72	76	73	93	103	125	131	138		
	PrecMax	1443	222	207	231	137	158	161	148	175	217	273	235	290		
	PrecMin	893	5	1	32	12	24	14	12	3	14	42	45	16		
	PrecStd	145	57	46	46	37	35	41	33	37	48	57	46	60		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	14.9	16.9	21.1	23.6	27.1	27.6	24.7	23.7	19.9	15.6	13.7		
		MCWB	11.6	10.0	11.2	13.8	16.8	18.7	19.1	19.0	18.2	15.8	14.0	12.4		
	2%	DB	11.8	11.8	14.1	17.2	20.6	23.2	24.4	22.2	21.0	17.5	14.5	13.0		
		MCWB	10.6	10.1	10.4	12.1	15.1	17.3	18.3	17.7	16.9	15.0	13.2	11.8		
	5%	DB	10.9	10.9	12.3	14.9	18.3	20.7	22.1	20.4	19.2	16.4	13.6	12.1		
		MCWB	9.6	9.4	9.9	11.0	14.1	16.0	17.4	16.8	15.9	14.4	12.3	10.9		
	10%	DB	10.1	10.1	11.1	13.2	16.3	19.0	20.2	19.2	18.0	15.4	12.7	11.1		
		MCWB	8.9	8.7	9.4	10.4	12.6	15.1	16.5	16.3	15.2	13.7	11.4	10.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	11.7	12.4	14.5	17.9	19.8	20.9	19.7	19.1	16.9	14.6	12.7		
		MCDB	12.7	13.0	14.9	19.9	22.3	25.1	25.3	23.3	22.0	18.2	15.4	13.3		
	2%	WB	11.0	10.6	11.2	12.7	15.9	17.7	19.0	18.4	17.6	15.6	13.5	12.0		
		MCDB	11.8	11.6	12.9	16.0	19.6	22.0	23.1	20.9	19.8	17.0	14.2	12.8		
	5%	WB	9.9	9.7	10.4	11.7	14.5	16.5	17.8	17.6	16.6	14.8	12.6	11.1		
		MCDB	10.7	10.7	11.9	14.1	17.5	19.9	21.2	19.6	18.5	16.0	13.4	11.9		
	10%	WB	9.0	8.8	9.6	10.7	13.3	15.6	16.9	16.8	15.7	13.9	11.5	10.1		
		MCDB	10.0	10.0	11.0	12.9	15.5	18.3	19.5	18.7	17.5	15.2	12.5	11.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	6.1	7.3	8.4	8.5	8.4	7.9	7.6	8.2	6.9	6.4	5.8		
		MCDBR	5.4	6.2	8.8	11.2	11.8	12.3	12.5	9.6	10.5	7.4	5.7	5.2		
		MCWBR	5.1	5.1	6.2	7.2	7.4	7.5	7.3	6.0	6.8	5.5	4.9	5.1		
	5% WB	MCDBR	5.3	5.6	7.6	9.7	10.4	10.7	10.3	8.2	8.7	6.5	5.3	5.1		
		MCWBR	5.3	5.0	6.0	6.6	7.0	6.9	6.7	5.6	6.2	5.4	5.0	5.2		
(40) Clear-Sky Solar Irradiance	taub		0.331	0.344	0.374	0.395	0.401	0.391	0.392	0.375	0.367	0.353	0.329	0.321		
	taud		2.292	2.337	2.268	2.240	2.269	2.327	2.332	2.393	2.408	2.402	2.354	2.272		
	Ebn at Noon		646	761	808	838	850	862	854	850	812	742	647	582		
	Edh at Noon		65	82	107	124	126	120	118	105	92	76	61	56		
(44) All-Sky Solar Radiation	RadAvg		0.72	1.41	2.58	4.03	5.01	5.39	4.95	4.10	3.05	1.71	0.87	0.55		
	RadStd		0.07	0.14	0.26	0.38	0.31	0.53	0.49	0.30	0.22	0.15	0.09	0.06		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page